

Free and Reduced Lunch Eligibility in US Public Schools

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Machine Learning

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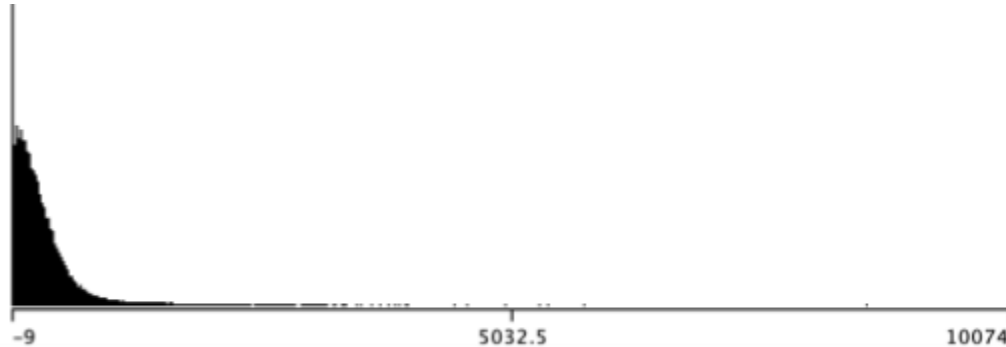
Goal

Our goal is to aid in the allocation of financial and dietary resources to schools who need them the most. We will be using multiple accessible data points to predict the number of students that are eligible for free and reduced-price lunch.

Original Dataset Description

We chose a federal dataset of public school characteristics, each of the 101,390 instances representing a public school. The dataset contains 75 attributes, and has a dimension of 74. For the purposes of this project, the class attribute is TOTFRL, the number of students that are eligible for free or reduced-price lunch. Across all attributes, the total number of missing values is 1,078,417. In addition, there are many impossible values such as -1 in multiple attributes. The name, description, and number of missing values of each attribute can be found in the appendix.

The class data is heavily skewed right, with a mean of 245.605 and a standard deviation of 286.334. The majority of these values lie close to zero. The data also has some outliers; the instance with the highest number of students eligible for free or reduced-price lunch has 10,074 such students.



Distribution of class data

Preprocessing

Cleaning

Issue 1: Irrelevant Attributes

This dataset contains many attributes that clearly cannot be used to predict the class label. To speed up preprocessing of attributes, we began with removing these attributes. The columns that were removed in this step are: OBJECTID, NCESSCH, SURVYEAR, STABR, LEAID, ST_LEAID, LEA_NAME, SCH_NAME, LSTREET1, LSTREET2, PHONE, LATCOD, and LONCOD. Attributes and their descriptions can be found in the appendix.

Issue 2: Repetitive Attributes

The grade attributes (PK, KG, G01-G13, AE, and UG) each have greater than 45% of values missing. The dataset also contains the school level, such as elementary or high school, and the total number of students at each school. SCHOOL_LEVEL and TOTAL summarize the data provided by the individual grade attributes. To simplify the dataset and sidestep the large proportion of missing values of the grade attributes, we removed the grade attributes, as well as the GSLO and GSHI.

Because we are predicting the total number of students eligible for free or reduced-price lunch, the attributes FRELCH and REDLCH would include the exact same data. For this reason, we removed these two attributes.

Issue 3: Special Values

According to the NCES, the organization that provided this dataset, the dataset contains multiple special values. -1 or M indicates missing values and -2 or N indicates the data is not applicable. -9 indicates that the data does not meet NCES standards. We treated all of these special values as missing values. We removed instances whose class labels were one of these three special values.

Issue 4: Missing Values

We first removed all instances with missing class labels, leaving us with a total of 86,637 instances. We began to address the missing values of other attributes by removing remaining attributes with greater than 40% of values missing. The attributes we removed in this step were: LZIP4 and DIRECTCERT. Next, we filled remaining missing values with mean (for numerical data) and mode (for nominal data).

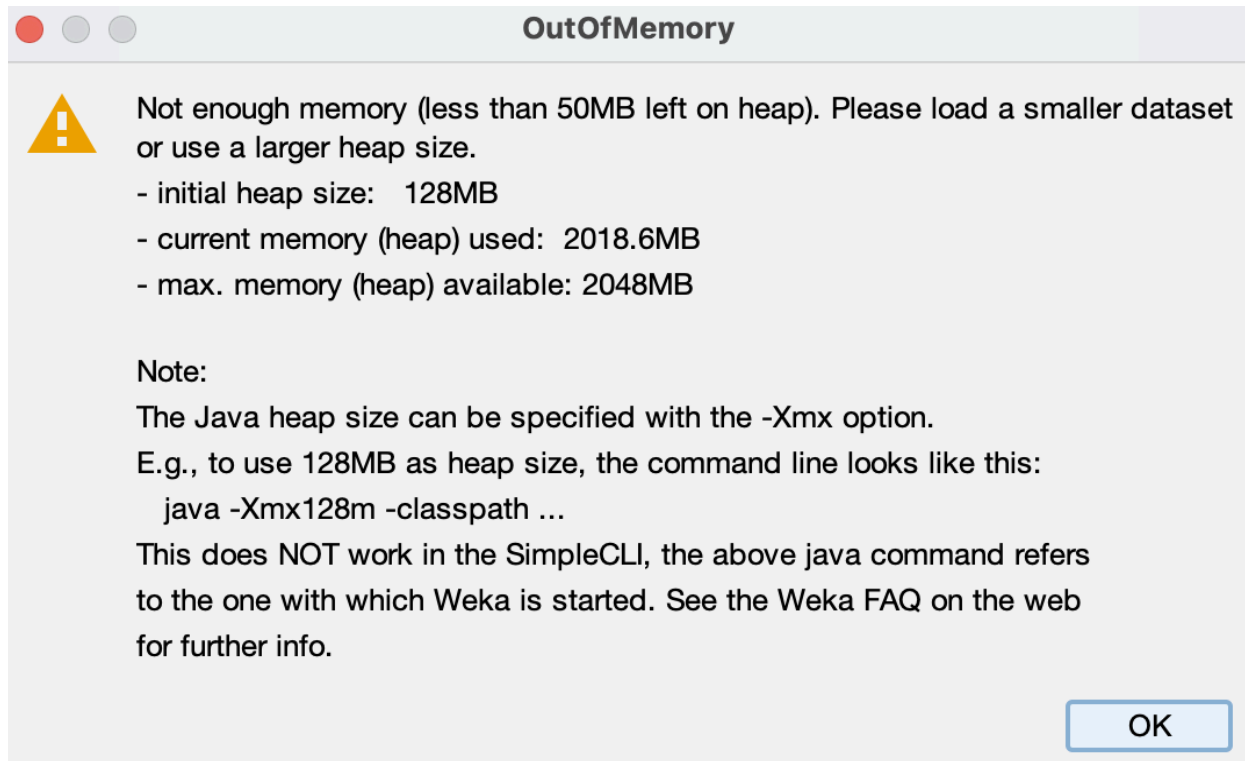
Issue 5: Wide range of existing class labels

The last issue we addressed in preprocessing is that the class data (number of students on a reduced lunch program) is numeric and discrete. We decided to split the class data into two groups: low, meaning that the number of students eligible for free or reduced-price lunch is below average; and high, meaning that the number of students eligible for free or reduced-price lunch is above average.

We considered converting the class values to the proportion of eligible students for free or reduced-price lunch. However, the goal of our project is to aid in the allocation of resources to schools. When deciding on the number of resources necessary to allocate to schools, the proportion of students eligible for free or reduced-price lunch is much less helpful than the number of these students. For this reason, we decided against converting the class data to a proportion.

Memory Issue

When attempting to run attribute selection algorithms on Weka, we were presented with an OutOfMemory error. To solve this issue, we took a random sample of 10% of our dataset to use for all future operations. The nature of our class labels (high being above average and low being below average) lead to a perfect 50/50 split of instances among these two possible labels. For this reason, we were able to use simple random selection to produce a sample with a distribution of classes that is equal to the distribution of the original dataset. This sample allowed us to run all tools without reaching maximum memory. This sample had a total of 8,664 instances.



Train/Test split

We decided to use 70% of the data as a training set, and the other 30% of the data as a test set. We split this test set in two, using 15% of our dataset (half of the test set) as a validation set, and using another 15% of our dataset (the other half of the test set) as a final test set. Each set has a distribution of class data representative of the dataset as a whole.

Tools Used

We used the Weka remove function to remove attributes. We used Excel's find and replace tools to replace special values with "?", then converted the Excel file to an .arff file using Weka's converter, setting the missing value field as "?". We used Weka's SubsetByExpression with the expression "not ismissing(ATT#)" to remove attributes with missing class labels. We used Weka's ReplaceMissing values filter to fill in missing values with mean (for numerical data) and mode (for nominal data). To take a sample of our dataset, we first used Weka's Randomize filter to randomize the dataset, then its RemovePercentage filter to remove 90% of the data. Finally, we used Weka's RemovePercentage filter to split the dataset into training and testing sets and to split the testing set into testing and validation sets. We ensured that the split datasets did not overlap by using each percentage once with and once without the InvertSelection field as true.

Attribute Selection

Correlation

We used Weka's correlation attribute evaluator with the ranker search method. The results were as follows. We used a cutoff value of 0.2; the attributes we removed were: SCHOOL_TYPE_TEXT, HPALM, HP, WHALM, WH, HPALF, WHALF, CHARTER_TEXT, AS_, ASALM, ASALF, ULOCALE, STUTERATIO, SY_STATUS_TEXT, LSTATE, STATUS, AMALM, AM, AMALF, VIRTUAL, SCHOOL_LEVEL, LZIP, NMCNTY, and LCITY. The

attributes with a correlation greater than 0.2 were: MEMBER, TOTAL, TOTMENROL, TOTFENROL, FTE, HI, HIALM, HIALF, BLALM, BL, BLALF, TR, TRALF, and TRALM. Weka's correlation attribute uses the Pearson correlation coefficient, calculated using equation (1), which is a measure of how strongly two variables correlate with each other linearly.

$$(1) \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

The results of CorrelationAttributeEval are as follows:

Attribute Evaluator

Choose **CorrelationAttributeEval**

Search Method

Choose **Ranker -T -1.7976931348623157E308 -N -1**

Attribute Selection Mode

☒ Use full training set

☐ Cross-validation

Folds: 10

Seed: 1

No class

Start Stop

Result list (right-click for options)

00:18:50 - Ranker + CorrelationAttributeEval

Attribute selection output

Attribute ranking.

Attribute Evaluator (supervised, Class (nominal): 39 TOTFRL):

Correlation Ranking Filter

Ranked attributes:

0.4589	15	MEMBER
0.4587	14	TOTAL
0.4574	12	TOTMENROL
0.4521	13	TOTFENROL
0.4444	16	FTE
0.4393	32	HI
0.4393	30	HIALM
0.4365	31	HIALF
0.3964	24	BLALM
0.3946	26	BL
0.3898	25	BLALF
0.2674	35	TR
0.2653	34	TRALF
0.2616	33	TRALM
0.1834	8	SCHOOL_TYPE_TEXT
0.0944	27	HPALM
0.0941	29	HP
0.0936	36	WHALM
0.0929	38	WH
0.0921	28	HPALF
0.0918	37	WHALF
0.0822	4	CHARTER_TEXT
0.0738	23	AS_
0.0738	21	ASALM
0.0734	22	ASALF
0.0709	10	ULOCAL
0.0633	17	STUTERATIO
0.058	9	SY_STATUS_TEXT
0.057	2	LSTATE
0.0562	7	STATUS
0.0467	18	AMALM
0.046	20	AM
0.0428	19	AMALF
0.038	5	VIRTUAL
0.0346	6	SCHOOL_LEVEL
0.023	3	LZIP
0.02	11	NMCNTY
0.0151	1	LCITY

Selected attributes: 15, 14, 12, 13, 16, 32, 30, 31, 24, 26, 25, 35, 34, 33, 8, 27, 29, 36, 38, 28, 37, 4, 23, 21, 22, 10, 17, 9, 2, 7, 18, 20, 19, 5, 6, 3, 11, 1 : 38

Status OK

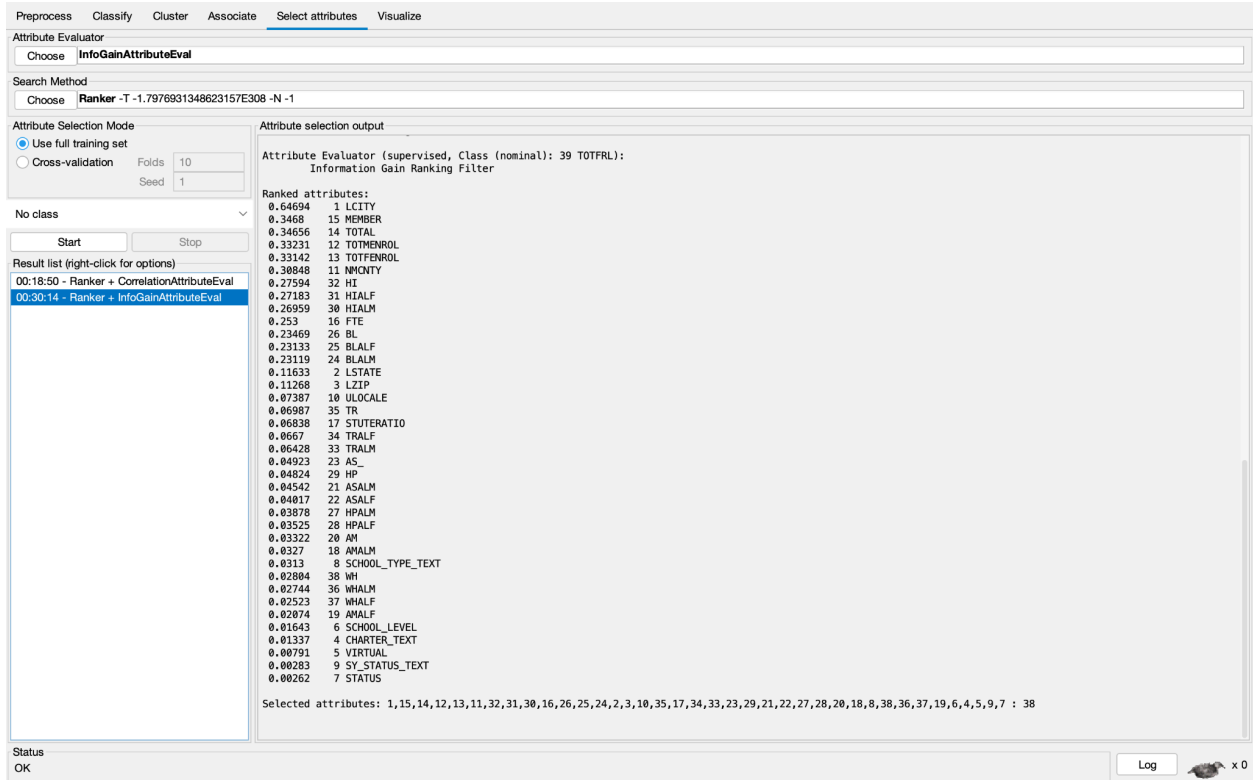
Log

Info gain

We used Weka's info gain attribute evaluator with the ranker search method. We used a cutoff value of 0.2; the attributes we removed were: LSTATE, LZIP, ULOCAL, TR, STUTERATIO, TRALF, TRALM, AS_, HP, ASALM, ASALF, HPALM, HPALF, AM, AMALM, SCHOOL_TYPE_TEXT, WH, WHALM, WHALF, AMALF, SCHOOL_LEVEL, CHARTER_TEXT, VIRTUAL, SY_STATUS_TEXT, and STATUS. The attributes with an info gain greater than 0.2 were: LCITY, MEMBER, TOTAL, TOTMENROL, TOTFENROL, NMCNTY, HI, HIALF, HIALM, FTE, BL, BLALF, and BLALM. Infogain is a measure of how well a variable reduces the entropy of another variable, calculated by equation (2).

$$(2) \text{InfoGain}(\text{Class}, \text{Attribute}) = H(\text{Class}) - H(\text{Class} | \text{Attribute}).$$

The results of InfoGainAttributeEval are as follows:

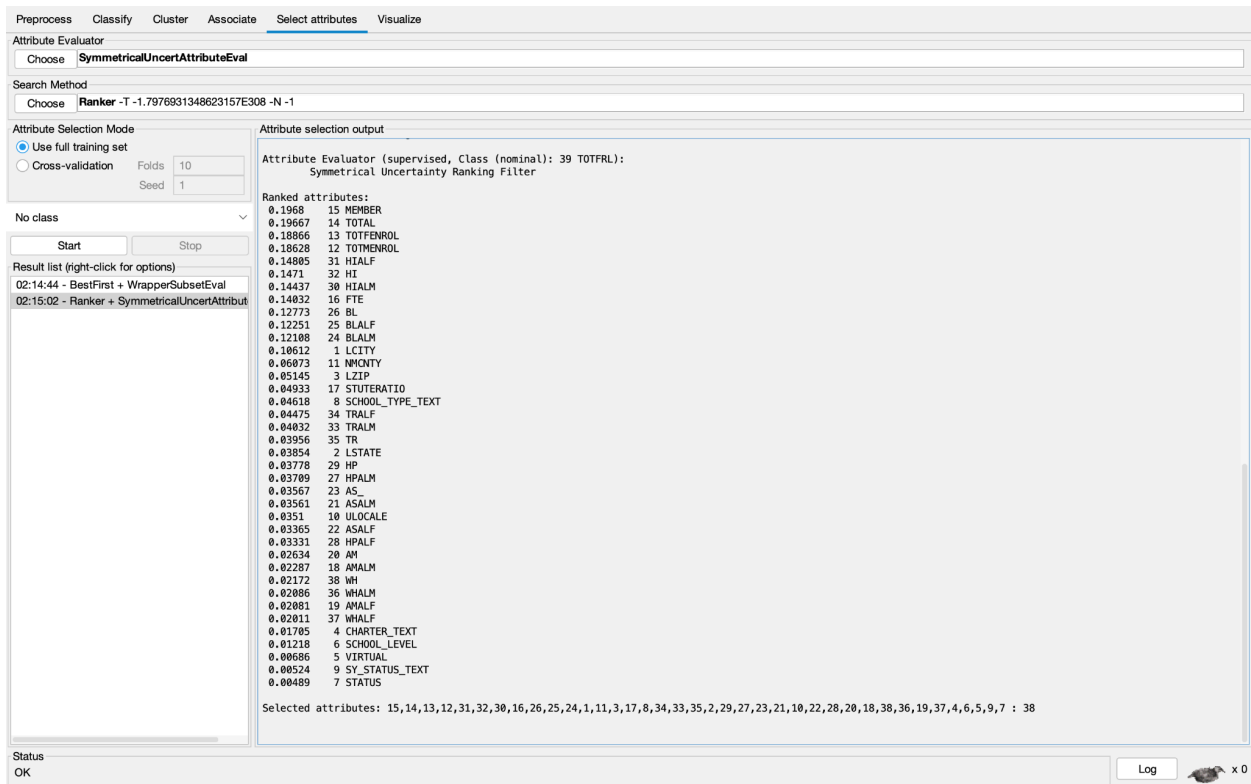


Symmetrical Uncertainty

We used Weka's symmetrical uncertainty attribute evaluator with the ranker search method. We used a cutoff value of 0.1; the attributes removed were: NMCNTY, LZIP, STUTERATIO, SCHOOL_TYPE_TEXT, TRALF, TRALM, TR, LSTATE, HP, HPALM, AS_, ASALM, ULOCAL, ASALF, HPALF, AM, AMALM, WH, WHALM, AMALF, AMALF, WHALF, CHARTER_TEXT, SCHOOL_LEVEL, VIRTUAL, SY_STATUS_TEXT, and STATUS. The attributes with a symmetrical uncertainty value greater than 0.1 were: MEMBER, TOTAL, TOTFENROL, TOTMENROL, HIALF, HI, HIALM, FTE, BL, BLALF, BLALM, and LCITY. Symmetrical uncertainty measures the symmetrical strength of dependency between variables calculated by equation (3).

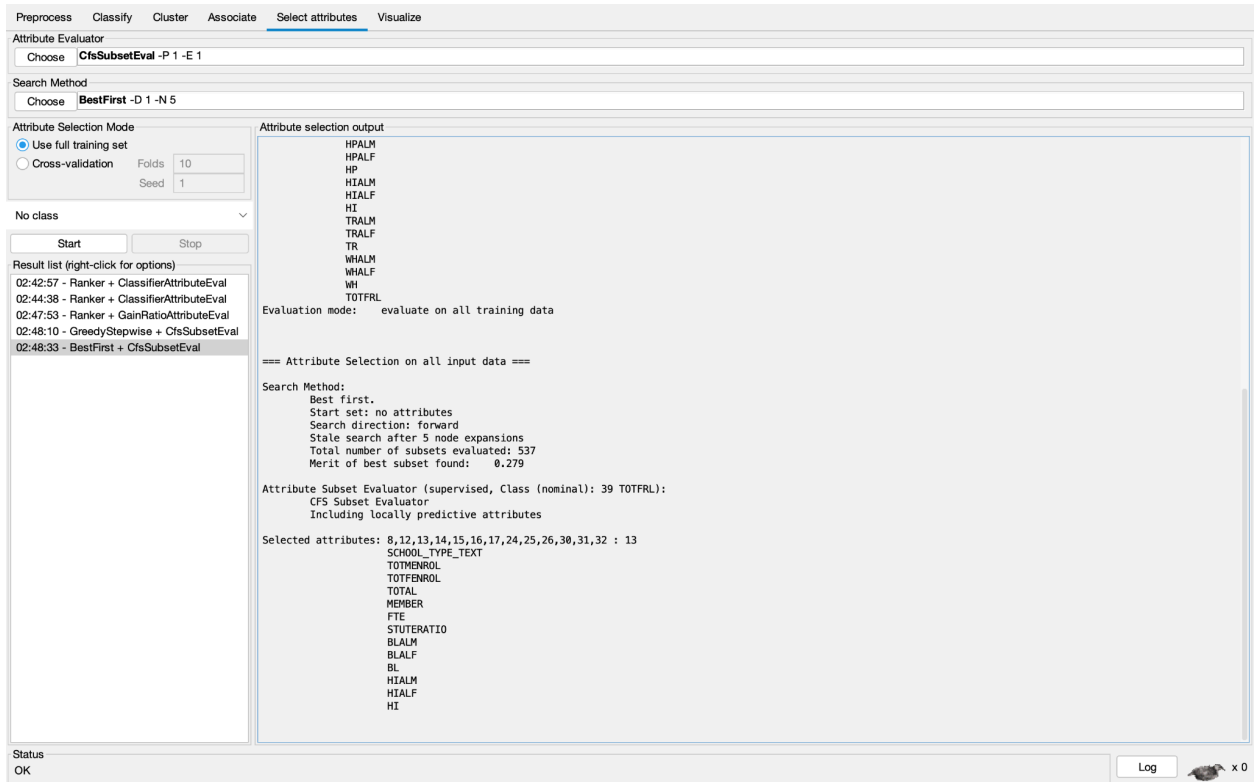
$$(3) \text{Symm}U(\text{Class}, \text{Attribute}) = \frac{2 * (H(\text{Class}) - H(\text{Class} | \text{Attribute}))}{H(\text{Class}) + H(\text{Attribute})}$$

The results of SymmetricalUncertAttributeEval are as follows:



Cfs Subset

We used Weka's cfs subset attribute evaluator with the best first search method. We chose to keep all attributes suggested by the evaluator. The attributes kept were: SCHOOL_TYPE_TEXT, TOTMENROL, TOTFENROL, TOTAL, MEMBER, FTE, STUTERATIO, BLALM, BLALF, BL, HIALM, HIALF, and HI. All other attributes were removed. CfsSubsetAttributeEval evaluates attributes by how correlated they are with the class, while also attempting to avoid selecting attributes highly correlated with each other. The results of CfsSubsetAttributeEval are as follows:



Intuition

Our final method of attribute selection was simply using our intuition. We considered how heavily we thought attributes might be related to the class, and picked the following attributes: LSTATE, LZIP, SCHOOL_LEVEL, ULOCALE, NMCITY, TOTAL, STUTERATIO, AM, AS_, BL, HP, HI, TR, and WH. All other attributes were removed.

Classification

We used the NaiveBayes, J48, 1R, and random tree classifiers. NaiveBayes uses probability to predict the class label assuming all attributes have an independent effect on the class. NaiveBayes simply calculates the probability of each class label and classifies the instance as whichever label had the highest probability. The J48 classification algorithm is a decision tree built using information entropy. The 1R classifier creates a rule to predict the output using a single single attribute. It uses only the values of this attribute to predict the class label. The random tree classifier creates multiple decision trees with random variables, and classifies each instance as the most frequently occurring tree output.

Results

CorrelationAttributeEval

NaiveBayes

Training accuracy was ~81%, validation accuracy was ~79%, testing results are as follows:

Summary

Correctly Classified Instances	1049	80.7544 %
Incorrectly Classified Instances	250	19.2456 %
Kappa statistic	0.6136	
Mean absolute error	0.189	
Root mean squared error	0.422	
Relative absolute error	37.7942 %	
Root relative squared error	84.4036 %	
Total Number of Instances	1299	

Detailed Accuracy By Class

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.921	0.310	0.754	0.921	0.829	0.630	0.916	0.912	'(-inf-217.5]'
	0.690	0.079	0.895	0.690	0.779	0.630	0.916	0.916	'(217.5-inf)'
Weighted Avg.	0.808	0.196	0.823	0.808	0.805	0.630	0.916	0.914	

Confusion Matrix

```

a   b   <-- classified as
608 52 |   a = '(-inf-217.5]'
```

```

198 441 |   b = '(217.5-inf)'
```

J48

Training accuracy was ~88%, validation accuracy was ~84%, testing results are as follows:

Summary

Correctly Classified Instances	1097	84.4496 %
Incorrectly Classified Instances	202	15.5504 %
Kappa statistic	0.6884	
Mean absolute error	0.2018	
Root mean squared error	0.3419	
Relative absolute error	40.3561 %	
Root relative squared error	68.3787 %	
Total Number of Instances	1299	

Detailed Accuracy By Class

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.897	0.210	0.815	0.897	0.854	0.692	0.907	0.899	'(-inf-217.5]'
	0.790	0.103	0.881	0.790	0.833	0.692	0.907	0.874	'(217.5-inf)'
Weighted Avg.	0.844	0.157	0.848	0.844	0.844	0.692	0.907	0.887	

Confusion Matrix

```

a   b   <-- classified as
592 68 |   a = '(-inf-217.5]'
```

```

134 505 |   b = '(217.5-inf)'
```

1R

Training accuracy was ~79%, validation accuracy was ~74%, testing results are as follows:

=== Summary ===

```

Correctly Classified Instances      961          73.98 %
Incorrectly Classified Instances    338          26.02 %
Kappa statistic                    0.4801
Mean absolute error                0.2602
Root mean squared error            0.5101
Relative absolute error             52.0435 %
Root relative squared error         102.0256 %
Total Number of Instances          1299

```

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.708	0.227	0.763	0.708	0.734	0.481	0.740	0.689	'(-inf-217.5]'
	0.773	0.292	0.719	0.773	0.745	0.481	0.740	0.668	'(217.5-inf)'
Weighted Avg.	0.740	0.259	0.741	0.740	0.740	0.481	0.740	0.678	

=== Confusion Matrix ===

```

  a   b  <-- classified as
467 193 |  a = '(-inf-217.5]
145 494 |  b = '(217.5-inf)

```

Random Tree

Training accuracy was ~100%, validation accuracy was ~80%, testing results are as follows:

=== Summary ===

```

Correctly Classified Instances      1035          79.6767 %
Incorrectly Classified Instances     264          20.3233 %
Kappa statistic                    0.5937
Mean absolute error                0.2032
Root mean squared error            0.4508
Relative absolute error             40.6494 %
Root relative squared error         90.168 %
Total Number of Instances          1299

```

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.780	0.186	0.812	0.780	0.796	0.594	0.797	0.745	'(-inf-217.5]'
	0.814	0.220	0.782	0.814	0.798	0.594	0.797	0.728	'(217.5-inf)'
Weighted Avg.	0.797	0.203	0.797	0.797	0.797	0.594	0.797	0.737	

=== Confusion Matrix ===

```

  a   b  <-- classified as
515 145 |  a = '(-inf-217.5]
119 520 |  b = '(217.5-inf)

```

InfoGainAttributeEval

NaiveBayes

Training accuracy was ~82%, validation accuracy was ~80%, testing results are as follows:

```

=== Summary ===
Correctly Classified Instances      1058          81.4473 %
Incorrectly Classified Instances    241          18.5527 %
Kappa statistic                    0.6274
Mean absolute error                0.1845
Root mean squared error            0.4157
Relative absolute error             36.8963 %
Root relative squared error        83.1537 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
                TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                0.932   0.307   0.758     0.932   0.836     0.645   0.918    0.916    '(-inf-217.5]'
                0.693   0.068   0.908     0.693   0.786     0.645   0.918    0.919    '(217.5-inf)'
Weighted Avg.    0.814   0.189   0.832     0.814   0.812     0.645   0.918    0.917

=== Confusion Matrix ===
   a   b  <-- classified as
 615  45 |   a = '(-inf-217.5]'
 196 443 |   b = '(217.5-inf)'

```

J48

Training accuracy was ~82%, validation accuracy was ~82%, testing results are as follows:

```

=== Summary ===
Correctly Classified Instances      1076          82.8329 %
Incorrectly Classified Instances    223          17.1671 %
Kappa statistic                    0.6552
Mean absolute error                0.2378
Root mean squared error            0.3459
Relative absolute error             47.5618 %
Root relative squared error        69.1805 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===
                TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                0.948   0.296   0.768     0.948   0.849     0.675   0.896    0.870    '(-inf-217.5]'
                0.704   0.052   0.930     0.704   0.801     0.675   0.896    0.859    '(217.5-inf)'
Weighted Avg.    0.828   0.176   0.848     0.828   0.826     0.675   0.896    0.864

=== Confusion Matrix ===
   a   b  <-- classified as
 626  34 |   a = '(-inf-217.5]'
 189 450 |   b = '(217.5-inf)'

```

1R

Training accuracy was ~86%, validation accuracy was ~63%, testing results are as follows:

```

=== Summary ===

Correctly Classified Instances      810          62.3557 %
Incorrectly Classified Instances    489          37.6443 %
Kappa statistic                    0.2433
Mean absolute error                0.3764
Root mean squared error            0.6135
Relative absolute error            75.2937 %
Root relative squared error        122.7171 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                0.776   0.534   0.600     0.776   0.677     0.255   0.621   0.580   '(-inf-217.5]'
                0.466   0.224   0.668     0.466   0.549     0.255   0.621   0.574   '(217.5-inf)'
Weighted Avg.   0.624   0.381   0.634     0.624   0.614     0.255   0.621   0.577

=== Confusion Matrix ===

  a    b  <-- classified as
512 148 |  a = '(-inf-217.5]'
341 298 |  b = '(217.5-inf)'

```

Random Tree

Training accuracy was ~100%, validation accuracy was ~76%, testing results are as follows:

```

=== Summary ===

Correctly Classified Instances      989          76.1355 %
Incorrectly Classified Instances    310          23.8645 %
Kappa statistic                    0.5242
Mean absolute error                0.2586
Root mean squared error            0.4368
Relative absolute error            51.717 %
Root relative squared error        87.3633 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                0.664   0.138   0.833     0.664   0.739     0.536   0.814   0.781   '(-inf-217.5]'
                0.862   0.336   0.713     0.862   0.780     0.536   0.814   0.759   '(217.5-inf)'
Weighted Avg.   0.761   0.235   0.774     0.761   0.759     0.536   0.814   0.770

=== Confusion Matrix ===

  a    b  <-- classified as
438 222 |  a = '(-inf-217.5]'
 88 551 |  b = '(217.5-inf)'

```

SymmetricalUncertAttributeEval

NaiveBayes

Training accuracy was ~81%, validation accuracy was ~80%, testing results are as follows:

=== Summary ===

Correctly Classified Instances	1061	81.6782 %
Incorrectly Classified Instances	238	18.3218 %
Kappa statistic	0.6321	
Mean absolute error	0.1842	
Root mean squared error	0.4155	
Relative absolute error	36.8419 %	
Root relative squared error	83.1044 %	
Total Number of Instances	1299	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.935	0.305	0.760	0.935	0.838	0.650	0.919	0.919	'(-inf-217.5]'
	0.695	0.065	0.912	0.695	0.789	0.650	0.919	0.919	'(217.5-inf)'
Weighted Avg.	0.817	0.187	0.835	0.817	0.814	0.650	0.919	0.919	

=== Confusion Matrix ===

```

a   b   <-- classified as
617 43 |   a = '(-inf-217.5]'
```

```

195 444 |   b = '(217.5-inf)'
```

J48

Training accuracy was ~83%, validation accuracy was ~82%, testing results are as follows:

=== Summary ===

Correctly Classified Instances	1079	83.0639 %
Incorrectly Classified Instances	220	16.9361 %
Kappa statistic	0.6598	
Mean absolute error	0.2326	
Root mean squared error	0.3433	
Relative absolute error	46.5263 %	
Root relative squared error	68.6575 %	
Total Number of Instances	1299	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.956	0.299	0.768	0.956	0.852	0.682	0.899	0.870	'(-inf-217.5]'
	0.701	0.044	0.939	0.701	0.803	0.682	0.899	0.867	'(217.5-inf)'
Weighted Avg.	0.831	0.173	0.852	0.831	0.828	0.682	0.899	0.868	

=== Confusion Matrix ===

```

a   b   <-- classified as
631 29 |   a = '(-inf-217.5]'
```

```

191 448 |   b = '(217.5-inf)'
```

IR

Training accuracy was ~86%, validation accuracy was ~64%, testing results are as follows:

```
=== Summary ===
```

```
Correctly Classified Instances      810          62.3557 %
Incorrectly Classified Instances    489          37.6443 %
Kappa statistic                     0.2433
Mean absolute error                 0.3764
Root mean squared error             0.6135
Relative absolute error             75.2937 %
Root relative squared error         122.7171 %
Total Number of Instances          1299
```

```
=== Detailed Accuracy By Class ===
```

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.776	0.534	0.600	0.776	0.677	0.255	0.621	0.580	'(-inf-217.5]'
	0.466	0.224	0.668	0.466	0.549	0.255	0.621	0.574	'(217.5-inf)'
Weighted Avg.	0.624	0.381	0.634	0.624	0.614	0.255	0.621	0.577	

```
=== Confusion Matrix ===
```

```
  a   b  <-- classified as
512 148 |  a = '(-inf-217.5]'
```

```
341 298 |  b = '(217.5-inf)'
```

Random Tree

Training accuracy was ~100%, validation accuracy was ~75%, testing results are as follows:

```
=== Summary ===
```

```
Correctly Classified Instances      959          73.826 %
Incorrectly Classified Instances    340          26.174 %
Kappa statistic                     0.4757
Mean absolute error                 0.3452
Root mean squared error             0.5029
Relative absolute error             69.0425 %
Root relative squared error         100.5791 %
Total Number of Instances          1299
```

```
=== Detailed Accuracy By Class ===
```

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.782	0.307	0.725	0.782	0.752	0.477	0.733	0.669	'(-inf-217.5]'
	0.693	0.218	0.755	0.693	0.723	0.477	0.733	0.692	'(217.5-inf)'
Weighted Avg.	0.738	0.263	0.739	0.738	0.738	0.477	0.733	0.680	

```
=== Confusion Matrix ===
```

```
  a   b  <-- classified as
516 144 |  a = '(-inf-217.5]'
```

```
196 443 |  b = '(217.5-inf)'
```

CfsSubsetAttributeEval

NaiveBayes

Training accuracy was ~81%, validation accuracy was ~80%, testing results are as follows:

```

=== Summary ===
Correctly Classified Instances      1058          81.4473 %
Incorrectly Classified Instances    241          18.5527 %
Kappa statistic                    0.6274
Mean absolute error                0.185
Root mean squared error            0.4168
Relative absolute error             36.9942 %
Root relative squared error        83.3643 %
Total Number of Instances          1299

=== Detailed Accuracy By Class ===
              TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
              0.933   0.308   0.758     0.933   0.836     0.646   0.921    0.923    '(-inf-217.5]'
              0.692   0.067   0.909     0.692   0.786     0.646   0.921    0.920    '(217.5-inf)'
Weighted Avg.   0.814   0.189   0.832     0.814   0.811     0.646   0.921    0.922

=== Confusion Matrix ===
   a   b   <-- classified as
616  44 |   a = '(-inf-217.5]'
197 442 |   b = '(217.5-inf)'

```

J48

Training accuracy was 88%, validation accuracy was ~84%, testing results are as follows:

```

=== Summary ===
Correctly Classified Instances      1092          84.0647 %
Incorrectly Classified Instances    207          15.9353 %
Kappa statistic                    0.6808
Mean absolute error                0.2002
Root mean squared error            0.3387
Relative absolute error             40.0443 %
Root relative squared error        67.7479 %
Total Number of Instances          1299

=== Detailed Accuracy By Class ===
              TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
              0.883   0.203   0.818     0.883   0.849     0.683   0.911    0.901    '(-inf-217.5]'
              0.797   0.117   0.869     0.797   0.831     0.683   0.911    0.881    '(217.5-inf)'
Weighted Avg.   0.841   0.161   0.843     0.841   0.840     0.683   0.911    0.891

=== Confusion Matrix ===
   a   b   <-- classified as
583  77 |   a = '(-inf-217.5]'
130 509 |   b = '(217.5-inf)'

```

1R

Training accuracy was ~79%, validation accuracy was ~74%, testing results are as follows:

=== Summary ===

Correctly Classified Instances	961	73.98 %
Incorrectly Classified Instances	338	26.02 %
Kappa statistic	0.4801	
Mean absolute error	0.2602	
Root mean squared error	0.5101	
Relative absolute error	52.0435 %	
Root relative squared error	102.0256 %	
Total Number of Instances	1299	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.708	0.227	0.763	0.708	0.734	0.481	0.740	0.689	'(-inf-217.5]'
	0.773	0.292	0.719	0.773	0.745	0.481	0.740	0.668	'(217.5-inf)'
Weighted Avg.	0.740	0.259	0.741	0.740	0.740	0.481	0.740	0.678	

=== Confusion Matrix ===

```

a   b   <-- classified as
467 193 |   a = '(-inf-217.5]'
```

```

145 494 |   b = '(217.5-inf)'
```

Random Tree

Training accuracy was ~100%, validation accuracy was ~80%, testing results are as follows:

=== Summary ===

Correctly Classified Instances	1029	79.2148 %
Incorrectly Classified Instances	270	20.7852 %
Kappa statistic	0.5842	
Mean absolute error	0.2079	
Root mean squared error	0.4559	
Relative absolute error	41.5732 %	
Root relative squared error	91.1869 %	
Total Number of Instances	1299	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.797	0.213	0.795	0.797	0.796	0.584	0.792	0.736	'(-inf-217.5]'
	0.787	0.203	0.790	0.787	0.788	0.584	0.792	0.726	'(217.5-inf)'
Weighted Avg.	0.792	0.208	0.792	0.792	0.792	0.584	0.792	0.731	

=== Confusion Matrix ===

```

a   b   <-- classified as
526 134 |   a = '(-inf-217.5]'
```

```

136 503 |   b = '(217.5-inf)'
```

Intuition

NaiveBayes

Training accuracy was ~80%, validation accuracy was ~80%, testing results are as follows:

=== Summary ===

Correctly Classified Instances	1044	80.3695 %
Incorrectly Classified Instances	255	19.6305 %
Kappa statistic	0.6059	
Mean absolute error	0.199	
Root mean squared error	0.4195	
Relative absolute error	39.807 %	
Root relative squared error	83.8957 %	
Total Number of Instances	1299	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.915	0.311	0.752	0.915	0.826	0.621	0.905	0.909	'(-inf-217.5]'
	0.689	0.085	0.887	0.689	0.775	0.621	0.905	0.898	'(217.5-inf)'
Weighted Avg.	0.804	0.200	0.819	0.804	0.801	0.621	0.905	0.903	

=== Confusion Matrix ===

```

a   b   <-- classified as
604 56 |   a = '(-inf-217.5]'
```

```

199 440 |   b = '(217.5-inf)'
```

J48

Training accuracy was ~84%, validation accuracy was ~84%, testing results are as follows:

=== Summary ===

Correctly Classified Instances	1078	82.9869 %
Incorrectly Classified Instances	221	17.0131 %
Kappa statistic	0.6587	
Mean absolute error	0.2271	
Root mean squared error	0.3424	
Relative absolute error	45.424 %	
Root relative squared error	68.481 %	
Total Number of Instances	1299	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.918	0.261	0.784	0.918	0.846	0.669	0.905	0.883	'(-inf-217.5]'
	0.739	0.082	0.897	0.739	0.810	0.669	0.905	0.875	'(217.5-inf)'
Weighted Avg.	0.830	0.173	0.840	0.830	0.828	0.669	0.905	0.879	

=== Confusion Matrix ===

```

a   b   <-- classified as
606 54 |   a = '(-inf-217.5]'
```

```

167 472 |   b = '(217.5-inf)'
```

1R

Training accuracy was ~79%, validation accuracy was ~74%, testing results are as follows:

```

=== Summary ===

Correctly Classified Instances      961           73.98 %
Incorrectly Classified Instances    338           26.02 %
Kappa statistic                    0.4801
Mean absolute error                0.2602
Root mean squared error            0.5101
Relative absolute error            52.0435 %
Root relative squared error        102.0256 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                0.708   0.227   0.763     0.708   0.734     0.481   0.740    0.689    '(-inf-217.5]'
                0.773   0.292   0.719     0.773   0.745     0.481   0.740    0.668    '(217.5-inf)'
Weighted Avg.    0.740   0.259   0.741     0.740   0.740     0.481   0.740    0.678

=== Confusion Matrix ===

  a    b  <-- classified as
467 193 |  a = '(-inf-217.5]'
145 494 |  b = '(217.5-inf)'

```

Random Tree

Training accuracy was ~100%, validation accuracy was ~75%, testing results are as follows:

```

=== Summary ===

Correctly Classified Instances      957           73.6721 %
Incorrectly Classified Instances    342           26.3279 %
Kappa statistic                    0.473
Mean absolute error                0.273
Root mean squared error            0.4998
Relative absolute error            54.608 %
Root relative squared error        99.9706 %
Total Number of Instances         1299

=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                0.761   0.288   0.732     0.761   0.746     0.473   0.746    0.694    '(-inf-217.5]'
                0.712   0.239   0.742     0.712   0.727     0.473   0.746    0.685    '(217.5-inf)'
Weighted Avg.    0.737   0.264   0.737     0.737   0.737     0.473   0.746    0.689

=== Confusion Matrix ===

  a    b  <-- classified as
502 158 |  a = '(-inf-217.5]'
184 455 |  b = '(217.5-inf)'

```

Analysis

From our results, we can see that CorrelationAttributeEval and CfsSubsetAttributeEval with J48 performed best with a testing accuracy of 84%. We can also see that CfsSubsetAttributeEval with NaiveBayes had the highest ROC area meaning that this particular classification model has a strong ability to distinguish between positive and negative classes. In the differences between the accuracy on the training and validation sets we can see that Random Trees in general tended to overfit the data resulting from a near %100 accuracy on the training dataset to a substantial difference on the validation dataset accuracy. Best Model in terms of F-score is J48 in both CorrelationAttributeEval and CfsSubsetAttributeEval with value of 0.84. So, Overall, the best model for our dataset would be either CorrelationAttributeEval with J48 or CfsSubsetAttributeEval with J48 because both perform very similarly and are the best classifiers when dealing with overfitting, underfitting, and F1 score.

Team Members and Tasks Performed

Finding the Data & Building Proposal: Lucas
 Preprocessing Initial Attempt: Parasmai
 Preprocessing & Project Update: Lucas
 Non-Weka Attribute Selection Algorithm: Lucas
 Attribute Selection Algorithms and Classifiers: Lucas
 Results Output: Lucas
 Results Analysis: Parasmai
 Building Final Report: Lucas
 Final Presentation: Parasmai

Appendix

Attributes

#	Name	Description	Missing Values
1	OBJECTID	instance number	0
2	NCESSCH	Unique School ID	0
3	SURVYEAR	Year corresponding to survey record	0
4	STABR	Postal state abbreviation code	0
5	LEAID	NCES Agency ID	0
6	ST_LEAID	State Local Education Number	0
7	LEA_NAME	Education Agency Name	0
8	SCH_NAME	School Name	0
9	LSTREET1	Location address, street 1	0
10	LSTREET2	Location address, street 2	100817
11	LCITY	Location city	0
12	LSTATE	Location state	0
13	LZIP	Location 5 digit ZIP code	0
14	LZIP4	Location Secondary ZIP code	42811
15	PHONE	Telephone number	0
16	CHARTER_TEXT	Whether a Charter School	0
17	VIRTUAL	Virtual School Status	0
18	GSLO	Grades Offered - Lowest	0
19	GSHI	Grades Offered - Highest	0
20	SCHOOL_LEVEL	School level	0
21	STATUS	Start of year status (code)	0
22	SCHOOL_TYPE_TEXT	School type (description)	0
23	SY_STATUS_TEXT	Start of year status (description)	0

24	ULOCAL	Locale Code	0
25	NMCNTY	County Name	0
26	FRELCH	Free Lunch Program	0
27	REDLCH	Reduced-Lunch Program	0
28	DIRECTCERT	Direct Certification	0
29	PK	Prekindergarten students	68998
30	KG	Kindergarten students	47329
31	G01	Grade 1 students	46978
32	G02	Grade 2 students	46921
33	G03	Grade 3 students	46931
34	G04	Grade 4 students	47132
35	G05	Grade 5 students	48376
36	G06	Grade 6 students	63367
37	G07	Grade 7 students	68166
38	G08	Grade 8 students	67898
39	G09	Grade 9 students	73289
40	G10	Grade 10 students	73501
41	G11	Grade 11 students	73502
42	G12	Grade 12 students	73574
43	G13	Grade 13 students	101257
44	UG	Ungraded students	93501
45	AE	Adult Education Students	101207
46	TOTMENROL	Total Male Enrollment	2480
47	TOTFENROL	Total Female Enrollment	2480
48	TOTAL	Total students	1671
49	MEMBER	Total elementary/secondary students (excludes adult enrollment)	1671
50	FTE	Total teachers	3853
51	STUTERATIO	Student-teacher ratio	1814
52	AMALM	American Indian/Alaska Native Male students	2581
53	AMALF	American Indian/Alaska Native Female students	2579
54	AM	American Indian/Alaska Native students	2533
55	ASALM	Asian students Male	2492
56	ASALF	Asian students Female	2490

57	AS_	Asian students	2484
58	BLALM	Black or African American students Male	2494
59	BLALF	Black or African American students Female	2497
60	BL	Black or African American students	2487
61	HPALM	Native Hawai'ian or Other Pacific Islander students Male	2608
62	HPALF	Native Hawai'ian or Other Pacific Islander students Female	2607
63	HP	Native Hawai'ian or Other Pacific Islander students	2561
64	HIALM	Hispanic students Male	2481
65	HIALF	Hispanic students Female	2480
66	HI	Hispanic students	2480
67	TRALM	Two or more races students Male	2487
68	TRALF	Two or more races students Female	2485
69	TR	Two or more races students	2484
70	WHALM	White students Male	2481
71	WHALF	White students Female	2481
72	WH	White students	2480
73	LATCOD	Latitude	0
74	LONCOD	Longitude	0
75	TOTFRL	Total of free lunch and reduced-price lunch eligible	0
	Total		1,078,417

Sources

Dataset: <https://catalog.data.gov/dataset/public-school-characteristics-current-340b1>